Chapter 5: Linked Lists

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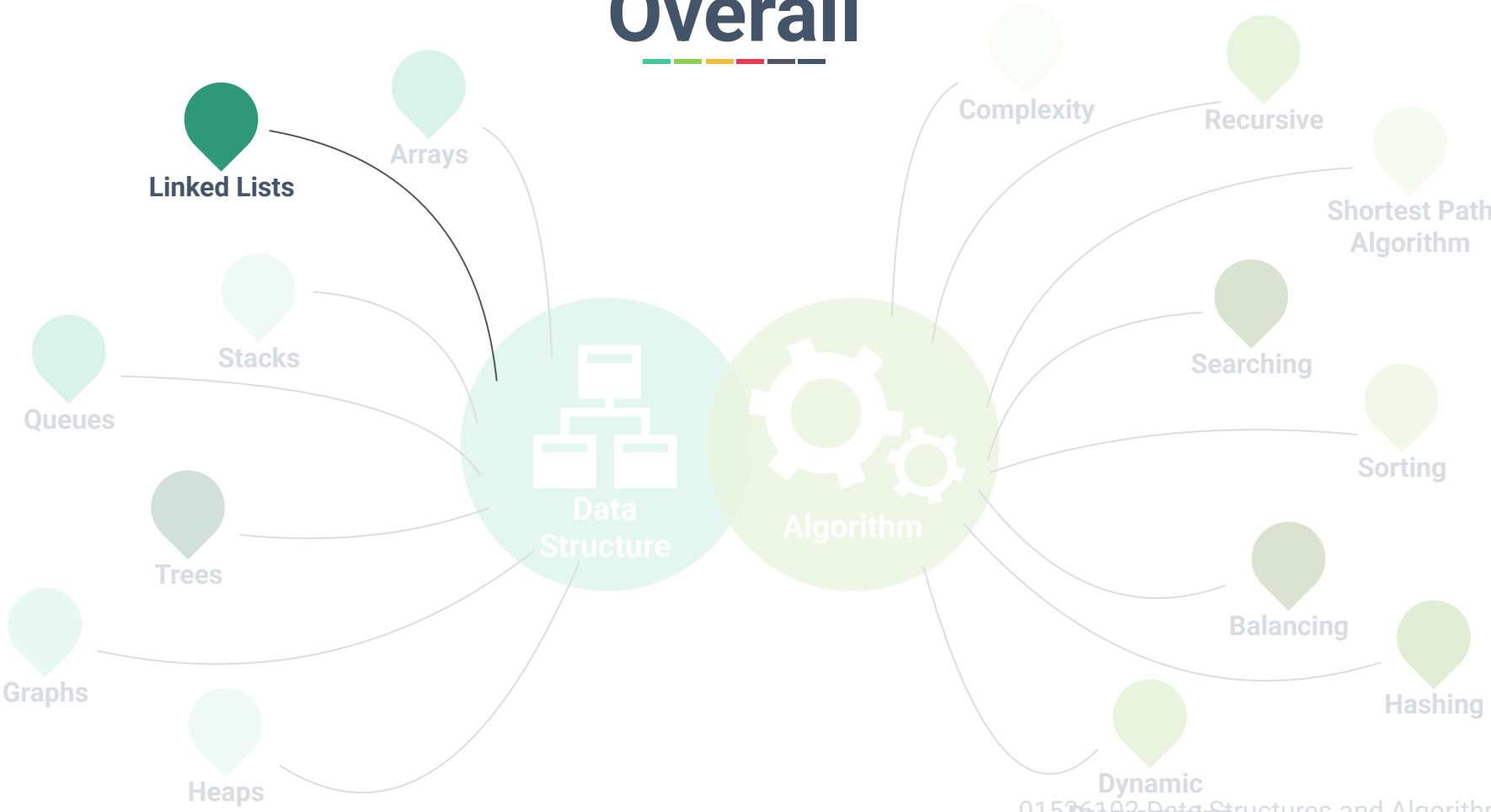
Overall


Linked Lists
Arrays
Stacks
Queues
Trees
Graphs
Heaps
Data Structure
Algorithm
Complexity
Recursive
Shortest Path Algorithm
Searching
Sorting
Balancing
Hashing
Dynamic Programming

Today's Outline

1. What is a Linked List?
2. Singly Linked Lists
    - Traversing
    - Insert a node
    - Delete a node

Previously

Python's array-based list

- Stack
- Queue

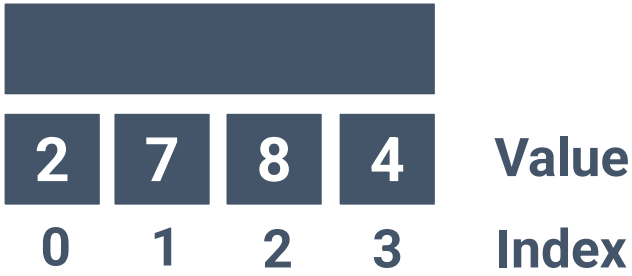
Disadvantages of array:

- Length of array has to be pre-allocated, empty space wasted.
- Adding or removing elements between values in the array is expensive - $O(n)$

Linked Lists

To avoid these limitations, an alternative to array is **linked list**.

Array



Linked lists

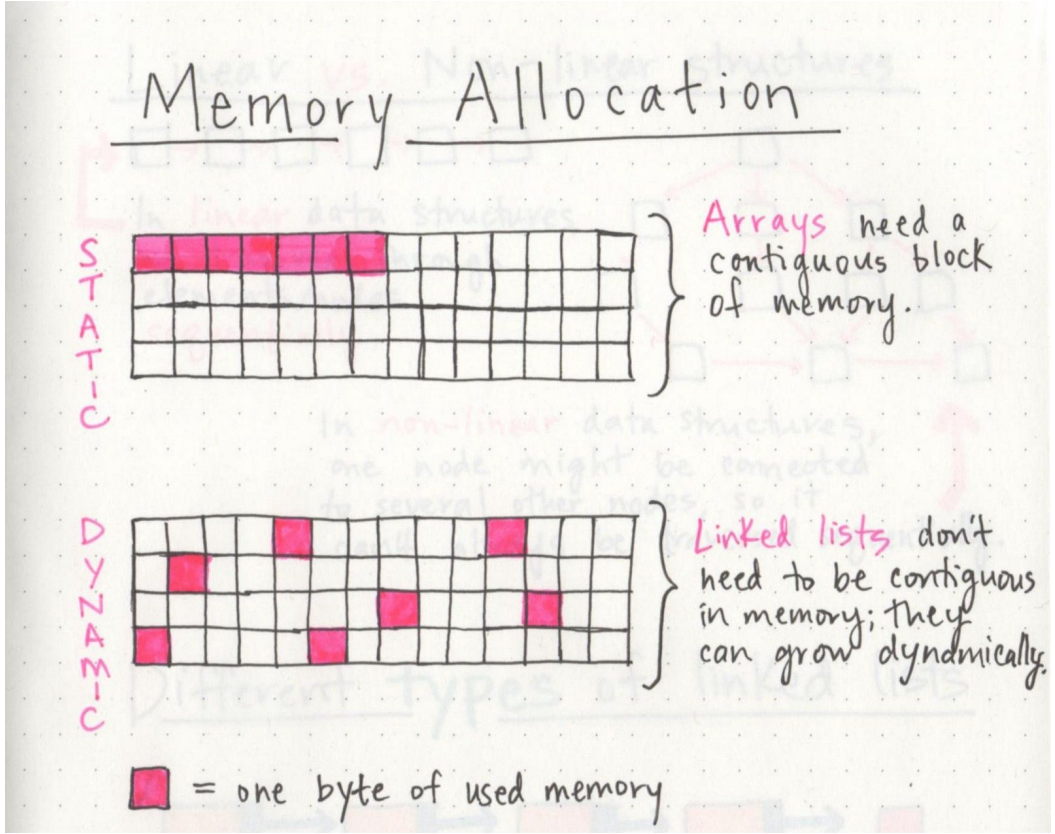
Linked Lists

Static

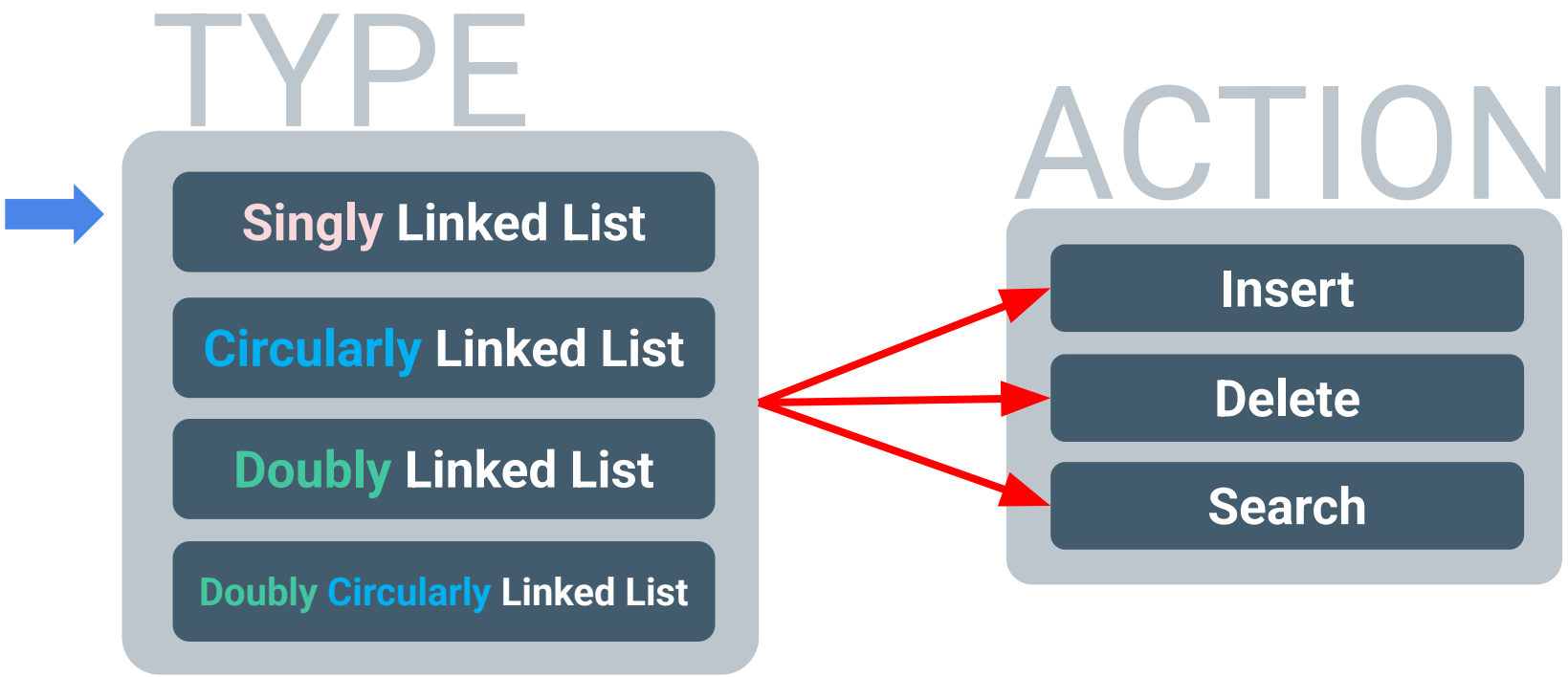
- Pre-allocated
- Fixed size
  - Unable to grow

Dynamic

- Allocated as needed
- Able to grow



Linked Lists



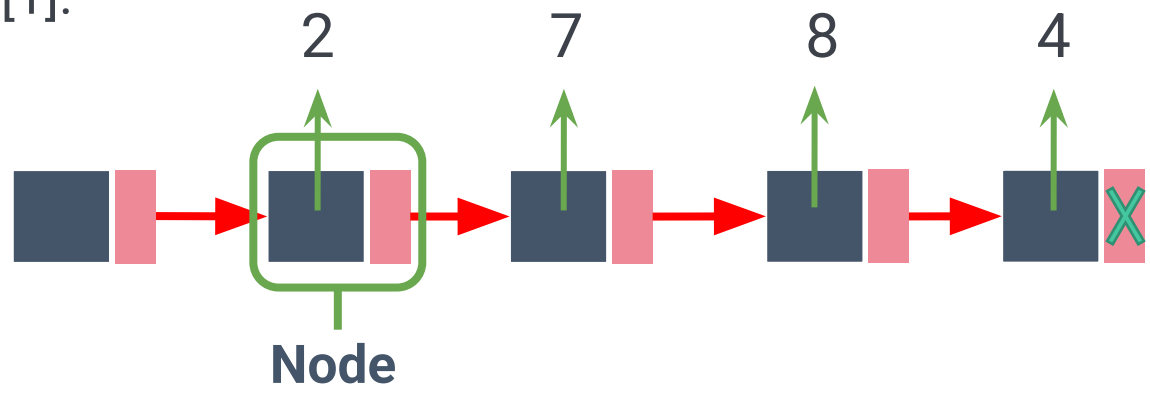
What is a Linked List?

A singly **linked list** is a collection of nodes that form a linear order of a sequence [1].

[1] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

What is a Linked List?

A singly **linked list** is a collection of nodes that form a linear order of a sequence [1].



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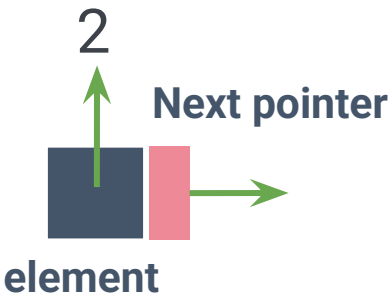
Linked List Node



Linked List Node Structures

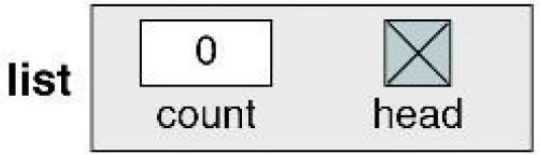
Create a Linked List

1. **Create a sentinel/header/root node**

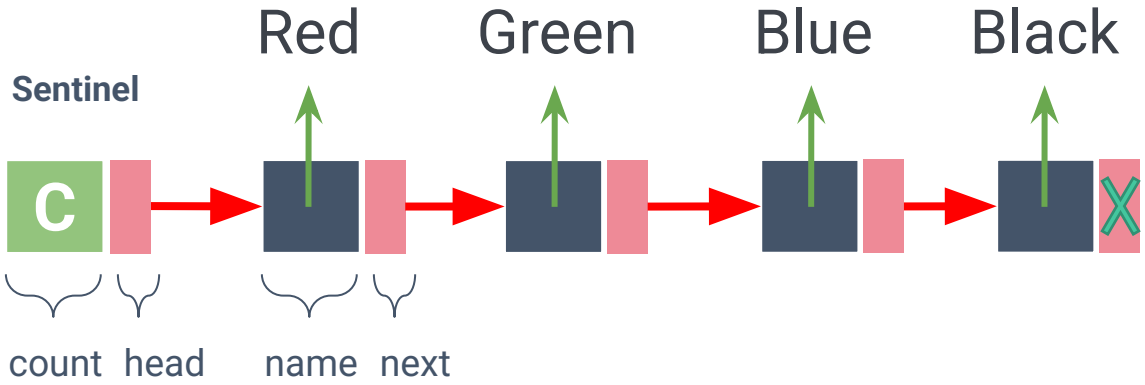
Algorithm createList (list)

1. Allocate a list
2. Set list head to null
3. Set list count to 0

End createList



Singly Linked Lists



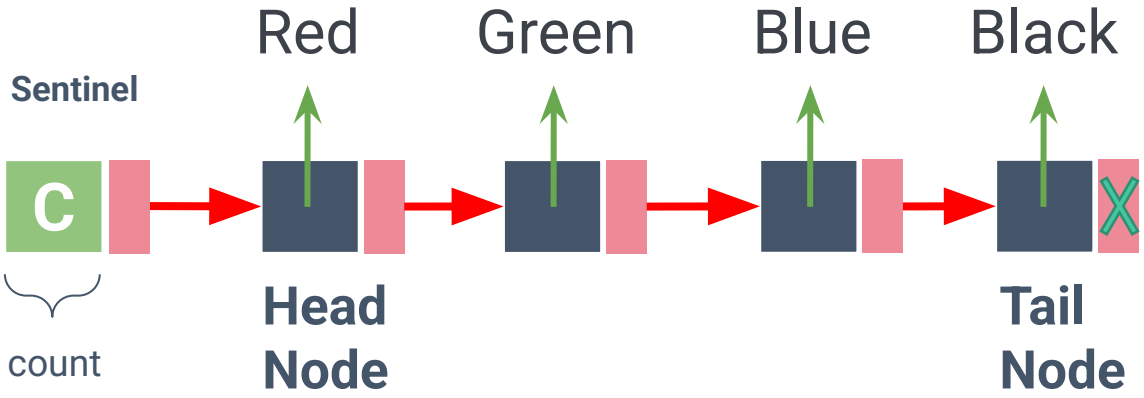
Color

String name

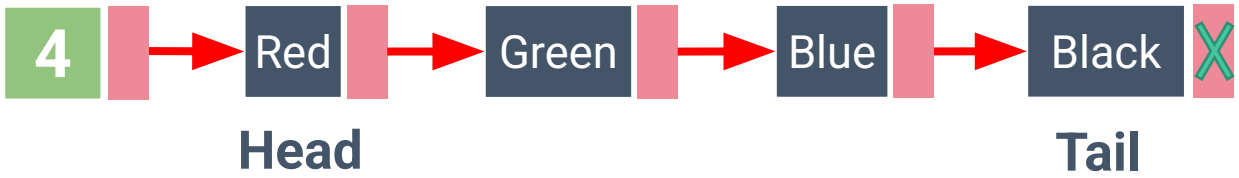
Color next

End Color

Singly Linked Lists



Singly Linked Lists



Create a Linked List

2. Create a data/element Node

```
Algorithm createDataNode (d, p)
    colorNew = allocate(Color)
    name = d
    next = p
    return colorNew
End createDataNode
```

Traversing Singly Linked Lists

Address/ Byte#	Value
6000	4
6001	6002
6002	2
6003	6008
6004	8
6005	6012
6006	
6007	
6008	7
6009	6004
6010	
6011	
6012	4
6013	None

Suppose that it takes 1 byte to store an integer.

Traversing Singly Linked Lists

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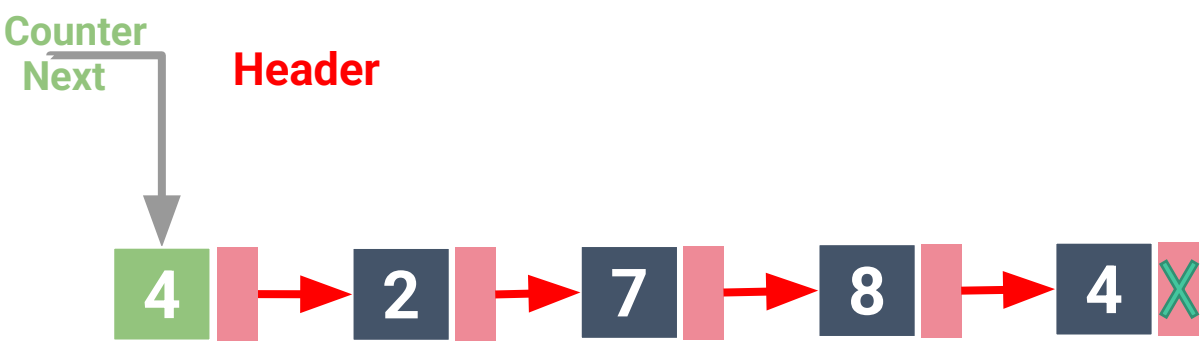

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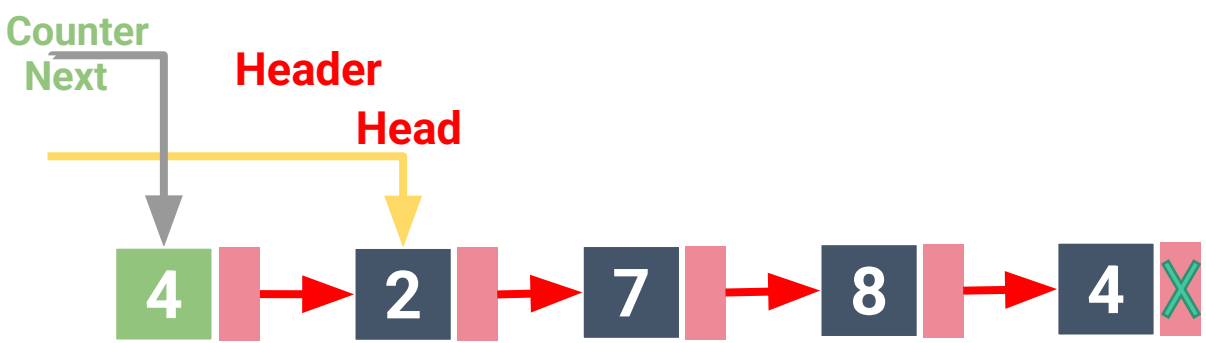

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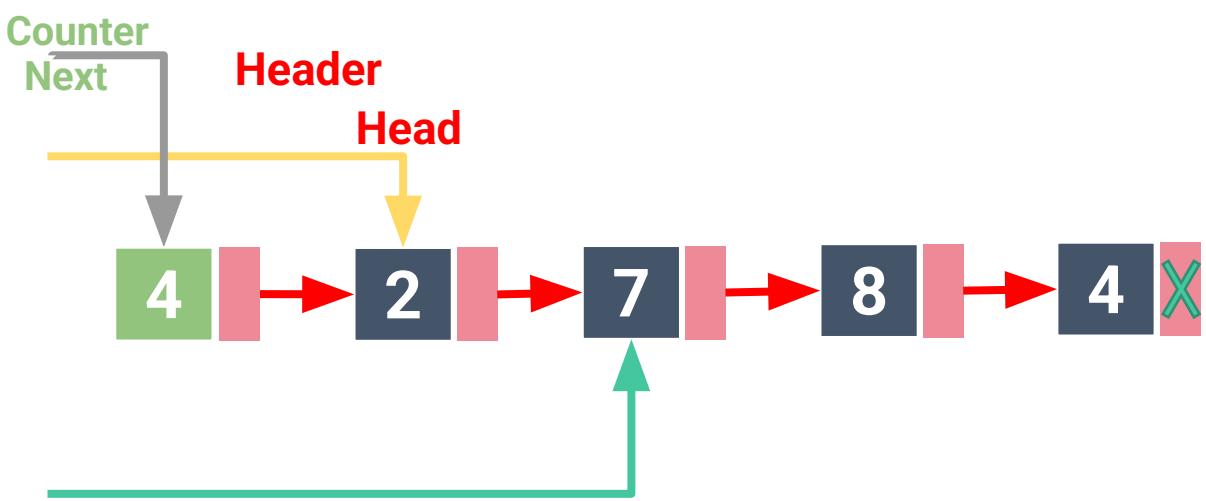

Traversing Singly Linked Lists

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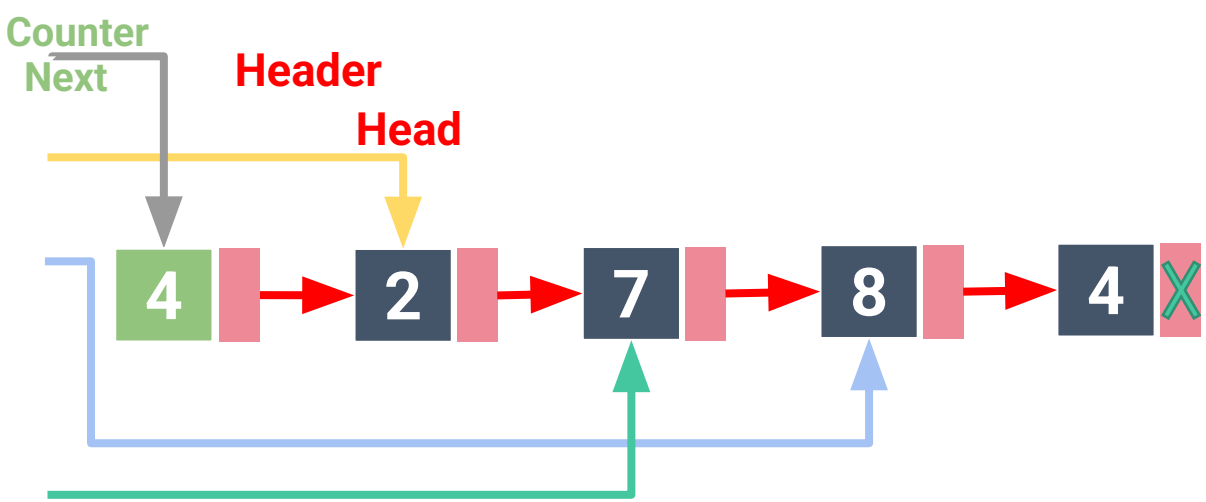
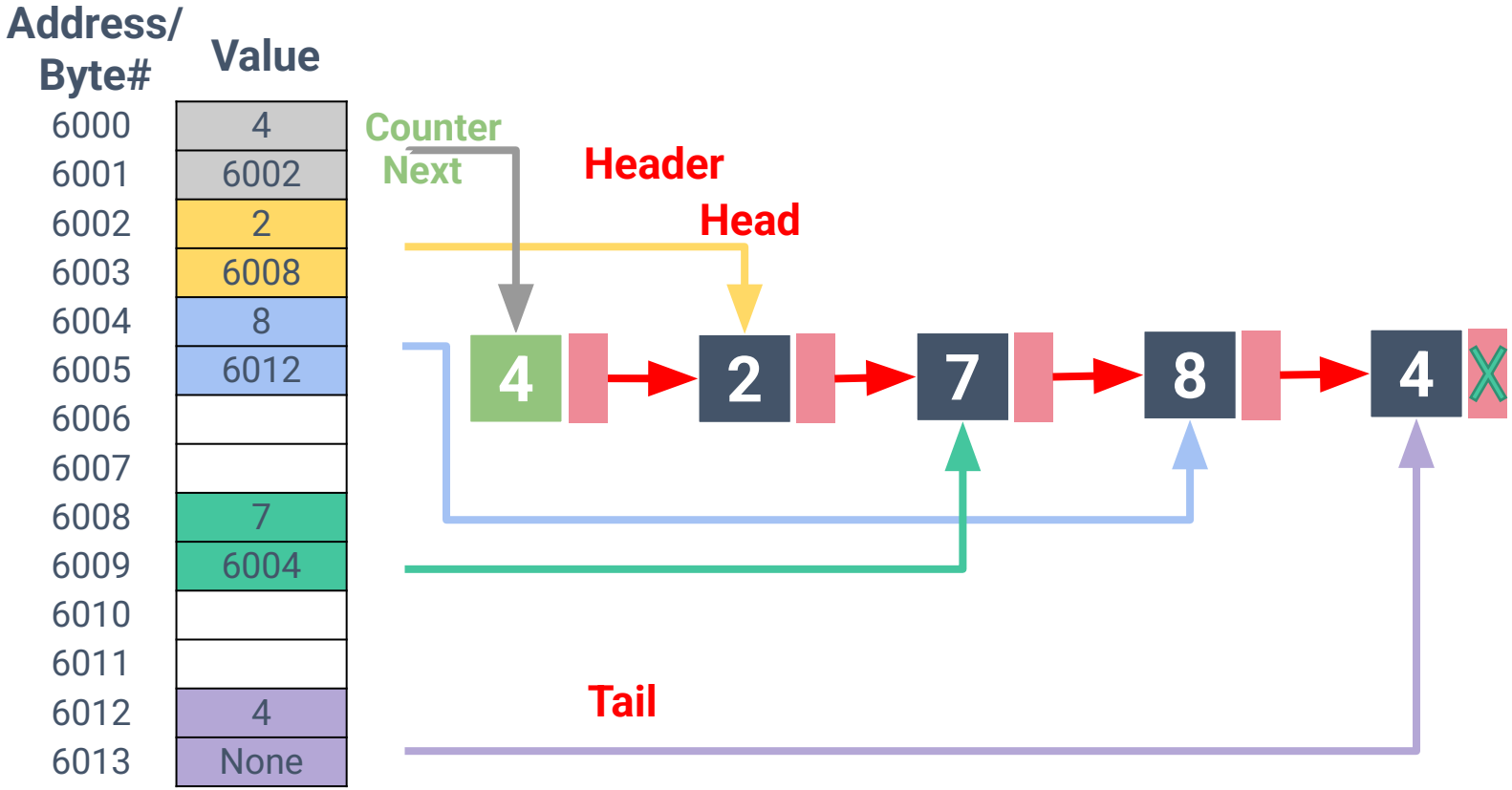
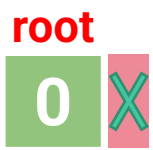


Traversing Singly Linked Lists



Insertion

Add “John” Node into a list



Insertion

at the tail of the list

Add “Tony” Node into a list

Insertion

at the tail of the list

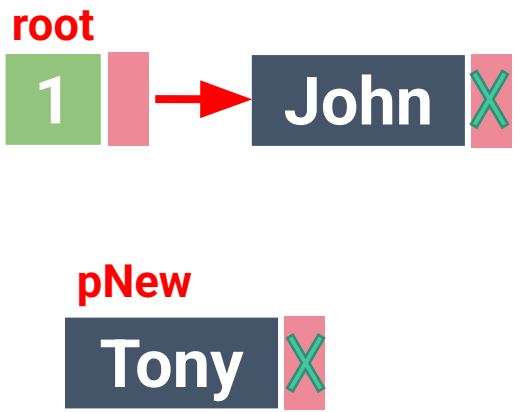
```
Algorithm add_last(L, e):
    new_node = Node(e)                  # Create new node instance
    new_node.next = None                # Set new node's next pointer to None
    if L.tail == None:                  # List was empty
        L.head & L.tail = new_node
    else:
        L.tail.next = new_node          # Make old tail point to new node
        L.tail = new_node               # Update the list's tail to reference the new node
    L.size = L.size + 1                 # Increment the node count
```

$O(1)$

Insertion

at the head of the list

Add “Tony” Node at the front of a list



Insertion

at the head of the list

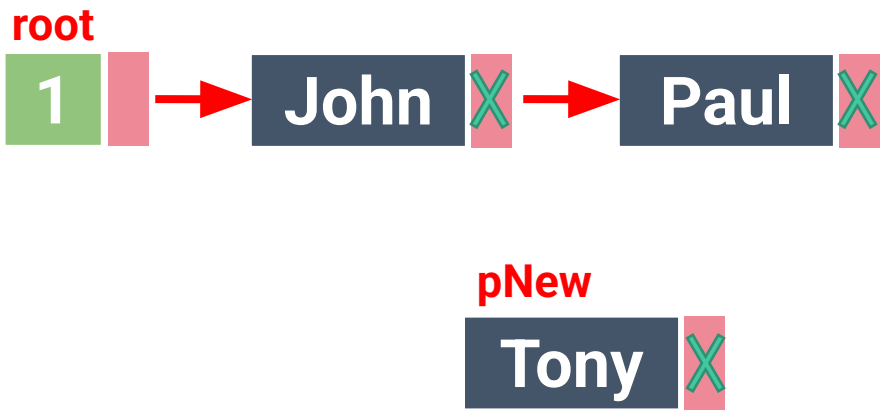
```
Algorithm add_front(L, e):
    new_node = Node(e)              # Create new node instance
    new_node.next = L.head          # Set new node's next pointer to the old head
    L.head = new_node               # Update the list's head to reference the new node
    L.size = L.size + 1             # Increment the node count
    if L.tail == None:              # List was empty
        L.tail = L.head
```

$O(1)$

Insertion

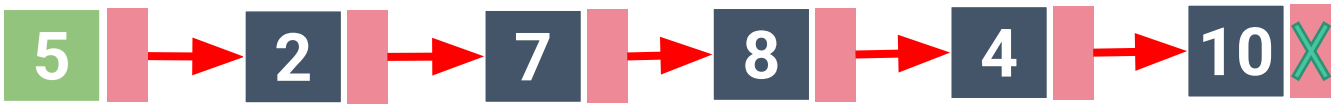
between nodes

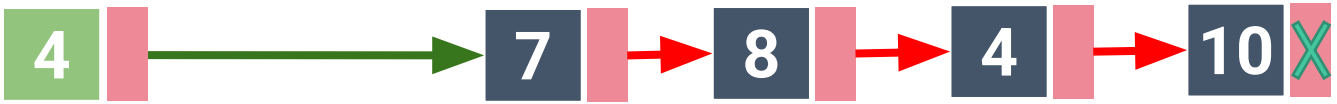
Add “Tony” Node between John and Paul



Deletion

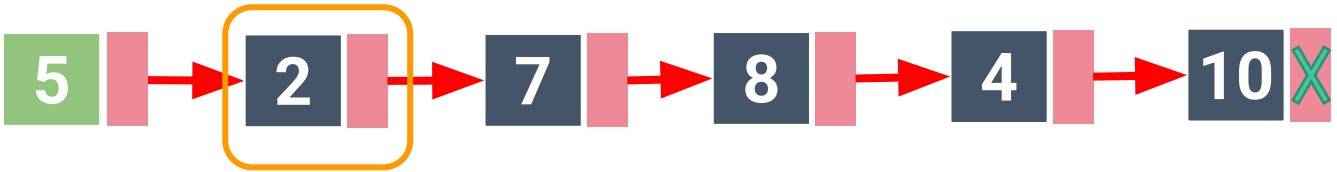
Delete the head node





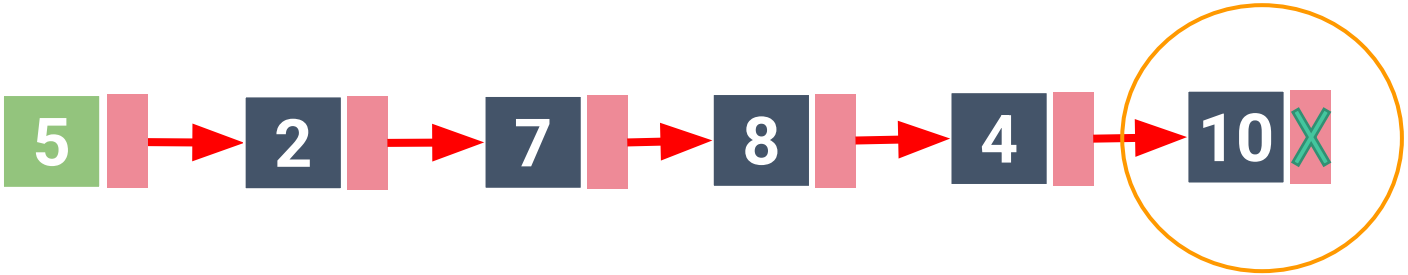
Deletion

Delete the head node



Deletion

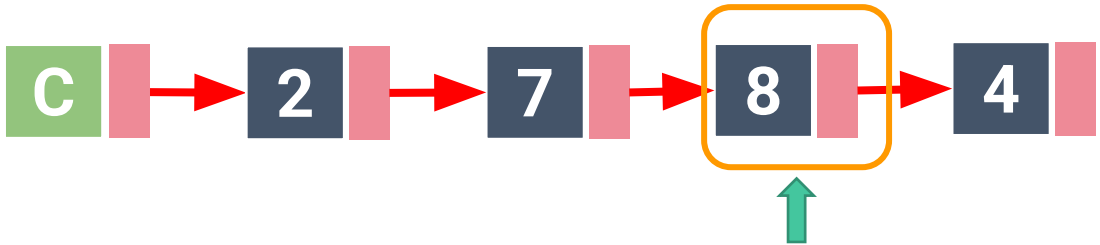
Delete the tail node



Deletion

Delete between nodes

Step 1



Step 2

Step 3